

A framework based on multivariate distribution-based virtual sample generation and DNN for predicting water quality with small data

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ABSTRACT

Deep Neural Network (DNN) is a powerful tool for predicting and monitoring water quality. However, its application is only limited to well-monitored zones where the availability of data for training and validation phases. In this study, we attempt to develop a novel framework based on Multivariate distributions (MVD) (elliptical copulas)-based Virtual Sample Generation (VSG) method to broaden the application of DNN to predict water quality even with a small dataset. This framework is evaluated to predict the Entropy Weighted Water Quality Index (EWQI) using DNN and Electrical Conductivity, Temperature, and pH as input variables, in Berrechid and Chaouia aquifer systems, Morocco. Validation results showed that the virtual samples generated from 400, 50, 30, and 20 original samples improved the *NSE* from 0.88 to 0.92, from 0.53 to 0.91, from 0.42 to 0.91, and from 0.24 to 0.87, respectively. Besides, sensitivity analysis of the methodology to the virtual data sizes and the original samples showed that the *RMSE* and *NSE* of the DNN models have limits in function to virtual data sizes according to the first order Exponential Decay and logistic trends, respectively. These limits highly depend on original sample sizes. Such empirical trends are crucial for reproducing the proposed methodology in other sites to determine optimal virtual datasets. Overall, the proposed methodology provided new insights to improve the DNN model performances in predicting water quality with small datasets. Hence, it is useful to manage water quality in order to supply clean water for the population in poorly monitored zones.

1. Introduction

Aquifer systems are vital strategic water resources for supplying water for drinking and irrigation purposes, especially in arid and semi-arid regions in which the availability of water surfaces is under high climatic variability (Wang et al., 2020; Zhai et al., 2022). In the past decades, the tremendous increase in pollution discharges due to anthropogenic activities combined with natural factors is deteriorating the water quality of the aquifers threatening the water suitability for drinking purposes (J. J. Huang et al., 2021; X. Yu X. Yu Huang et al., 2021; Zhang et al., 2021). However, accurate and reliable prediction models have become powerful tools to cheaply and early evaluate water resources' quality (Abba et al., 2021b; El Bilali et al., 2020a, 2020b; Zhao et al., 2022).

For three decades, two main categories of water quality models have been developed and explored by agencies and researchers, namely: process-based models and data-based models. For instance, MODFLOW is one of the powerful models for predicting groundwater quality

(Sullivan et al., 2019; Wei et al., 2019; Wei and Bailey, 2021). As for the water surfaces, US Environmental Protection Agency (USEPA) developed BASINS software which embeds several models and packages to predict water quality, such as SWAT, HSPF, and PLOAD models (Cheng et al., 2021; Lee et al., 2020; Xingpo et al., 2021). These models are based on the chemical transport mechanism of pollutants. Therefore, their application is only limited in well-monitored zones, especially the availability of physical parameters for calibration and validation processes (Oliver et al., 2016). Whereas, ML models rely on archived datasets for training and tuning processes rather than explaining mechanisms involved in the system to be modeled. Yet, it showed high accuracy prediction in water quality prediction e.g. (Castrillo and García, 2020; Chou et al., 2018; Ezzaoui et al., 2022; Wang et al., 2017). Therefore, data-based models could be an alternative in predicting water quality. Deep Neural Network (DNN) has been recognized as a powerful tool in modeling non-linear and complex systems. Many hydrologists are poised to explore the capability of DNN in various fields of water resource planning and management (Costache et al., 2022; Liu et al.,

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2022; Wang et al., 2022; Xiaoying Yang et al., 2022). For example, Bennett and Nijssen (2021) coupled DNN into process-based hydrologic models for predicting turbulent heat fluxes and found that the developed approach outperformed traditional process-based models. Salloom et al. (2021) developed a DNN model to now-cast the water demand and found that the error was decreased by about 30%. Similarly, DNN models also showed high accuracy in predicting water quality (Prasad et al., 2022; Yu et al., 2022; Zhou, 2020). Additionally, ensemble DNN (ENS-DNN) models have provided good accuracy compared to stand-alone ones in several studies because it is capable to reduce the variance and enhance the model stability (Asakura et al., 2018; Kumari and Toshniwal, 2021; Zhou et al., 2022). Interestingly, most studies showed that the DNN models outperformed process-based models and traditional machine learning models. Hence, it is a need to improve their application in predicting water quality for different conditions, especially in small data circumstances.

Indeed, DNN modeling requires large datasets for the training process, which is among the main challenges in its application since it is not always easy to get sufficient data, particularly in developing countries. Therefore, the application of DNN in predicting water quality appears difficult in poorly monitored zones. In this instance, virtual datasets could be valuable alternatives in a scarce data environment in applying DNN. Therefore, developing and exploring new virtual data generation methods to overcome the data shortage issues and difficulties in applying DNN models to predict water quality.

Several virtual data generation methods have been developed and explored by researchers. These methods can be classified into three classes, namely: 1) feature extraction-based method, 2) accumulating generator operator (AGO) based on the gray theory e.g. Box Gray Model (1,1) (BGM(1,1)) (Chang et al., 2015), Gray Bootstrap method (Wang et al., 2014), and 3) Virtual Sample Generation (VSG) method (Niyogi et al., 1998). Feature extraction-based methods showed their suitability to reduce the dimensionality of missing observed datasets in medicine studies (Espezuza et al., 2015; Zhang et al., 2019). Meanwhile, Accumulating Generator Operator and Virtual Sample Generation are outstanding approaches for generating virtual datasets (Abdul Lateh et al., 2017; Yang et al., 2011). They allow generating great size virtual datasets from small original samples. For instance, Chen et al. (2017) proposed a Particle Swarm Optimization-based VSG method to generate virtual data for enhancing the prediction accuracy of the Extreme Learning Machine (ELM) in chemical industries. They found that generated virtual datasets fairly improved ELM performances, with Error Improvement Rates (EIR) reaching 34.28%. Similarly, Gong et al. (2017) proposed a Monte Carlo and PSO-based VSG method for applying ELM models with small datasets to enhance the prediction accuracy of the energy consumption, in petrochemical industries. El Bilali et al. (2021a) proposed an integrated approach based on Gaussian noise data augmentation and AdaBoost models to predict fecal coliforms in rivers using small datasets. Also, other VSG-based methods were proposed and applied for different purposes, particularly in industrial fields e.g. (Chang et al., 2014; He et al., 2018; Liu and Li, 2020; MacAllister et al., 2020; Olesen and Shaker, 2021; Zhu et al., 2020, 2021a, 2021b). Interestingly, all these methods showed a potential improvement of ML models for regression problems by generating virtual data from small original samples. As for the DNN-based VSG, the generative adversarial network (GAN) is the most popular method (Goodfellow et al., 2014). In the study of Ledig et al. (2017), the authors found that the GAN method is useful to improve the image resolution from a down-sampled image. As for the regression tasks, Xu et al. (2020) developed an approach based on the Random Standard Sampling and DNN to predict the volatile fatty acid production during the anaerobic fermentation process. The last method was evaluated by Yin et al. (2022) and confirmed the potential accuracy improvement of DNN with small datasets compared to the baseline model trained with raw original samples. However, VSG methods are under continuous exploration, and their applicability differs from one field to another and depends on the dimensionality of the

variables.

As for water quality modeling, the Entropy Weighted Quality Index (EWQI) has shown promise in terms of subjectivity reduction in assessing water quality (Adimalla et al., 2019; Hossain and Patra, 2020; Salman et al., 2021). Moreover, accurate DNN models could be useful in developing software sensors to predict the EWQI in real-time to early detect the contamination risk (Arndt et al., 2022). Such promise technologies justify developing Virtual Sample Generation (VSG) methods to broaden the application of DNN to predict water quality in sparse data conditions, especially in developing countries, where a lot of poorly monitored zones (Saad Abdelkarim, 2020). However, generating appropriate virtual samples is a great challenge in VSG methods. Especially, the EWQI depends on various combinations of chemical parameters and factors. On the other hand, it is not recommended to use all chemical parameters as input variables due to the cost of the laboratory analysis and because their use will dramatically decrease field practical implications. In this instance, firstly generating appropriate virtual combinations of all variables respecting interdependences between these variables, and then selecting relevant inputs are necessary to avoid the loss of physical information.

However, the aforementioned methods rely on baseline models developed with small datasets to generate the output variables. Indeed, the output depends on several combinations of variables for example EWQI; using baseline models could generate inappropriate virtual data. Additionally, the sensitivity analysis of the VSG methods to the original samples is not explored in the literature even if it is required for practical implications. Also, the reproducibility of existed VSG methods in other cases is not further discussed and is not clear. To address the issue related to generating appropriate virtual combinations of datasets, it is crucial to develop and explore a new VSG to improve the accuracy and application of DNN models to water quality modeling with small datasets. Integrating either Multivariate Distribution (MVD) or copulas methods and using the most involved variables in the proposed VSG could be valuable to generate appropriate variable combinations in virtual datasets. To that end, new VSG methods should be explored and validated with enough experimental data to broaden the application of DNN in water resources studies.

This study is an attempt to develop a new method called MVD-VSG to improve the accuracy of DNN models with small dataset circumstances to predict EWQI in a poorly monitored zone. The proposed methodology was validated with enough experimental datasets to test its robustness. Furthermore, the sensitivity analysis of the MVD-VSG to the original samples and virtual data sizes was conducted to determine the optimal virtual dataset to be generated for a given small dataset. Finally, the reproducibility of the methodology in real engineering cases was clarified and discussed.

However, the rest of the paper is organized as follows: Section 2 describes the water quality challenges in the study area and describes the modeling framework; Section 3 presents the evaluation results of the proposed methodology while section 4 presents the discussion of the results, the practical implication, and future works, the conclusion of this research is presented in the last section.

2. Material and methods

2.1. Study area description and data

The aquifer systems studied are Berrechid and Chaouia aquifer systems with a global surface area of about 2700 Km² (Fig. 1). These water resources in the region supply water for drinking, industrial, agricultural, and touristic purposes in rural zones. Climatically, this region is a semi-arid environment with a rainfall ranging from 280 mm. yr⁻¹ to 320 mm. yr⁻¹, and more than 90% of the precipitation is received between October to April, while the temperature ranges from 6.5 °C (January) to 38 °C (August). Such conditions combined with socio-economic activities lead the management of these aquifers under

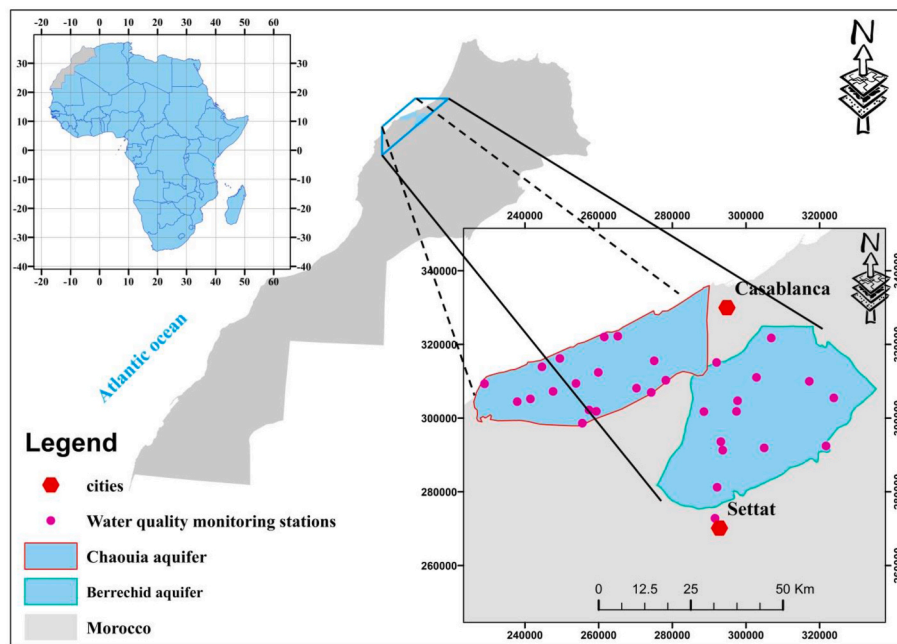


Fig. 1. Study area mapping, in Morocco.

several constraints, including water quality degradation, over-exploitation, and continuous increase in water demand. The water quality in this area is monitored by 31 monitoring station with a density of 87 Km² per station (Fig. 1). While the water management processes for agricultural purposes have been extensively studied (El Bilali et al., 2022, 2021b, 2020b; FAO, 2020; Ouassissou et al., 2019), water quality for drinking purposes has not taken much attention even the aquifers are crucial for livelihood of the isolated rural communities. The prediction approach could be useful to improve the monitoring and assessing water quality for drinking purposes. Therefore, this study aims at exploring DNN to predict EWQI with high accuracy even with small dataset that could be useful for new wells.

2.2. Dataset used

Datasets of 750 samples were recorded at 31 water quality monitoring stations (Fig. 1). The used datasets were provided by the River Basin Agency of Bouregreg and Chaouia (ABHBC). Sampling protocol was carried out two times during spring and the summer for the period from 2008 to 2019. Indeed, measurements concern the Electrical conductivity (EC), pH, Temperature, Cl⁻, SO₄²⁻, HCO₃⁻, NO₃⁻, Na⁺, K⁺, Ca²⁺, and Mg²⁺ concentrations. Physical parameters were measured in situ during the sampling. While, other parameters were analyzed in the laboratory through a flame photometer for analyzing Sodium and Potassium concentrations, a UV spectrophotometer for analyzing nitrate and Sulfate concentrations, and volumetric titration methods for the other ion concentrations. However, Table 1 presents statistical characteristics of raw datasets provided by ABHBC.

Table 1
Statistical characteristics of raw datasets (n = 750 samples), provided by ABHBC.

	Cl ⁻	NO ₃ ⁻	HCO ₃ ⁻	SO ₄ ²⁻	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	EC	Te	pH
Max	5063.00	717.00	3325.00	1440.00	2015.00	97.80	846.00	601.00	13000.00	41.50	9.30
Mean	868.53	90.51	329.94	147.98	413.55	5.04	174.90	92.73	3164.38	22.02	7.35
Min	87	8	50	64	50	0.06	6.10	0.02	754	7.15	4.10
SD	780.83	86.17	221.02	143.06	346.35	7.12	119.04	83.91	2225.12	3.72	0.40

2.3. Modeling framework description

In this subsection, we presented the proposed method called MVD-VSG to improve the prediction accuracy of the Entropy Weighed water Quality Index (EWQI) of aquifer systems by generating virtual data to train DNN models. Indeed, the EWQI is generally used to reduce subjectivity in evaluating water quality. Herein, we are tackling two challenges, which are developing DNN models with small datasets and using only physical parameters as inputs to overcome the data shortage and to increase the field practical implication of the DNN, respectively.

2.3.1. Entropy Weighted water Quality Index (EWQI)

The information entropy was introduced for the first time in thermodynamic fields (Shannon, 1948). It describes the information degree of uncertainty involved in the forecasting output of a given probabilistic event and can identify an amount of pertinent information from disorderly datasets. The entropy method was applied in various hydrological fields, such as drought indices, water quality, and flood risk assessment (Al-Hinai and Abdalla, 2021; Chang et al., 2016; Salman et al., 2021). Therefore, the information entropy was considered in this study to reduce the subjectivity in predicting groundwater quality.

For evaluating a water quality using $j = \{1, 2, 3 \dots n\}$ parameters and $i = \{1, 2, 3 \dots m\}$ water samples using the information entropy.

$$X_{ij} = \begin{matrix} & x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{matrix} \quad (1)$$

where m is the total number of the water samples taken and n is the total

number of the water quality parameters used for the evaluation process.

To reduce the dimensionality and unity effects of parameters the normalization of datasets is required. Indeed, this process can be conducted according to either the cost type or efficiency type of construction functions. For the cost type, the construction function is as follows:

$$y_{(ij)} = \frac{x_{\max(i,j)} - x_{(ij)}}{x_{\max(i,j)} - x_{\min(j,j)}} \quad (2)$$

As for the efficiency type, the construction function of the normalization is given by equation (3).

$$y_{(ij)} = \frac{x_{(ij)} - x_{\min(i,j)}}{x_{\max(i,j)} - x_{\min(j,j)}} \quad (3)$$

Hence,

$$Y_{ij} = \begin{matrix} y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & y_{22} & \dots & y_{2n} \\ y_{m1} & y_{m2} & \dots & y_{mn} \end{matrix} \quad (4)$$

where $y_{(ij) \in [0,1]}$ are normalized datasets represented by standard-grade matrix Y_{ij} .

Then the ratio P_{ij} of j index value in i sample

$$P_{ij} = \frac{y_{ij}}{\sum_i y_{ij}} \quad (5)$$

Hence, the information entropy e_j is calculated by the following equation.

$$e_j = -\frac{1}{\ln(m)} \sum_{i=1}^m P_{ij} \ln(P_{ij}) \quad \text{where} \quad \lim_{P_{ij} \rightarrow 0} P_{ij} \ln(P_{ij}) = 0 \quad (6)$$

Then the entropy weight ω_j is calculated by the following equation.

$$\omega_j = \frac{1 - e_j}{\sum_{i=1}^m (1 - e_j)} \quad (7)$$

The rating quality scale q_j of parameter j is calculated with the following equation:

$$q_j = \frac{C_j}{S_j} * 100 \quad (8)$$

where C_j is the concentration of the parameter j in mg.l^{-1} and S_j is the limit of the parameter j for water suitability for drinking purposes according to the World Health Organization (WHO) standards. If the parameter j is totally absent C_j is 0, then $q_j = 0$. Finally, the Entropy Weighted Water Quality Index (EWQI) is calculated by equation (9).

$$EWQI = \sum_{j=1}^n \omega_j * q_j \quad (9)$$

2.3.2. Multivariate distribution-virtual sample generation (MVD-VSG) method

The EWQI depends on several chemical parameters, which are inter-correlated, especially relationships between cations and anions due to various hydro-geochemical facies (El Ghali et al., 2020). Therefore, integrating these combinations is crucial in generating appropriate virtual datasets. MVD-VSG methodology is based on the multivariate distribution (MVD) to generate virtual datasets to overcome the data shortage issue and DNN to predict the EWQI.

2.3.2.1. Multivariate normal distribution (MVN). The Multivariate Normal (MVN) distribution of a random vector $X = \{x_1, x_2, x_3 \dots x_p\}$ with dimension p ($X \sim N(\mu, \Sigma)$) is characterized by its mean vector $\mu = \{\mu_1, \mu_2, \mu_3 \dots \mu_p\}$ and covariance matrix $\Sigma = (\Sigma_{ij})_{p \times p}$ describes the interdependence between random variables and is a symmetric positive definite (SPD) matrix ($p \times p$). Also, each random variable x_i has its distribution $N \sim (\mu_i, \Sigma_{ii})$.

The function density of the MVN can be written as follows:

$$f_X(x) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right) \quad x \in \mathbb{R}^p \quad (10)$$

Accordingly, if data have multivariate normal distribution implies that each variable has a normal distribution, but the opposite is not always true. However, the MVN can generate appropriate combinations of virtual datasets using the simple Monte Carlo method through Cholesky decomposition, as it takes into account the interdependences between variables.

2.3.2.2. Copulas. Generating appropriate virtual datasets using MVN is conditioned by the normality test. Since in a scarce dataset environment, the normality test related to MVN could be influenced either by data size or by the fact that the variables obey different distributions. However, the copulas are a promising alternative to generating appropriate combined virtual variables (Alidoost et al., 2019; Wang et al., 2019). These methods are a multivariate cumulative distribution function (CDF) where the marginal probability distribution of variables is uniform $[0,1]$ (Sklar, 1959). They are useful to separate the interdependence of the multivariate random vector from its marginal distribution.

Let $X \in \mathbb{R}^p$, the random vector considered. Its joint distribution function can be written as:

$$H(x_1, x_2, \dots, x_p) = C(F_1(x_1), F_2(x_2), \dots, F_p(x_p)) \quad X^T \in \mathbb{R}^p \quad (11)$$

where $C: [0,1]^p \rightarrow [0,1]$ is the copula and F_i denotes the i th marginal distribution function.

By turning equation (11) around, we can project any MVD to the unit square to recover the copula C .

$$C(u_1, u_2, \dots, u_p) = H(F_1^{-1}(u_1), F_2^{-1}(u_2), \dots, F_p^{-1}(u_p)) \quad (12)$$

where F_i^{-1} is the inverse marginal distribution function of F_i .

Indeed, the copulas can be divided into two classes, such as elliptical and Archimedean methods. Archimedean copula families are powerful to model bivariate distributions because of their simplicity and properties (Chang et al., 2016; Wang et al., 2019; X. Xiaoying Yang et al., 2022). In this study, we adopted elliptical copulas as they can generate and simulate random variables with high dimensions and are widely used in hydrological sciences (Tootoonchi et al., 2020). The projection of the MVN induces the Gaussian Copula with a density given by equation 13 (Arbenz, 2013).

$$C_r(u) = \frac{1}{\sqrt{|r|}} \exp\left(-\frac{1}{2} F_i^{-1T} (r^{-1} - I) F_i^{-1}\right) \quad (13)$$

where $r \in [-1, 1]^{p \times p}$ is the correlation matrix with 1 in diagonal considered and I is identity matrix.

2.3.3. Deep Neural Network (DNN)

Deep Neural Network (DNN) as one of the ML algorithms is based on Artificial Neural Networks (ANN) with considerable numbers of hidden layers (Rumelhart et al., 1985). Through various changeable structures, the DNN approach is capable to map the complexity and non-linearity of the system to be modeled. The weighted sums of the inputs are computed in each neuron and pass to the next through a function transfer (Haykin, 1999). In this study, sigmoid and linear activation functions were selected in Hidden layers and the output layer, sequentially. The learning process aim is to adapt the hyper-parameters of the networks until the actual output correspond to the predicted value using optimization algorithm and cost function. However, in this study, the Levenberg-Marquardt algorithm and mean square error (MSE) as cost function were employed.

2.3.4. Method and implementation

In this study, first the raw dataset was pre-processed to check the

data quality through the Charge Balance Error (CBE) water samples, where those which have CBE of more than 5% in absolute value should be deleted (Freeze et al., 1979). Then, an exploratory data analysis (EDA) of the cleaned data was carried out. The EDA includes the normality tests and correlation matrix through visualization analysis. The cleaned data (D_s) was randomly divided into the training (D_{train}) and validation ($D_{validation}$) dataset.

Statistics and Machine Learning toolbox in Matlab (R2021b) was used to implement the proposed methodology. From the training dataset D_{train} , small datasets of 50, 30, and 20 samples were randomly sampled without replacement (generally dataset with sample size less than 50 is considered small data). These datasets were normalized in a range of [0, 1] to reduce the dimensionality effects. Then, a series of virtual datasets with samples ranging from 500 to 10,000 samples were generated from D_{train} , 50, 30, and 20 samples according to either MVN or Gaussian copula. After virtual data generation through MVD-VSG method, a series of DNN models were constructed using the Electrical Conductivity (EC), pH, and Temperature as input variables to predict EWQI. Indeed, the DNN structures play a crucial role in the improvement of prediction accuracy. Hence, we evaluate various DNN structures by changing neuron numbers (from 5 to 10) and hidden layer numbers (from 3 to 6), learning rate, epoch numbers, and activation functions, and then we selected the best structures. The developed DNN were called to simulate the EWQI related to $D_{validation}$ to evaluate the DNN model performances. However, Fig. 2 summarizes the proposed methodology MVD-VSG.

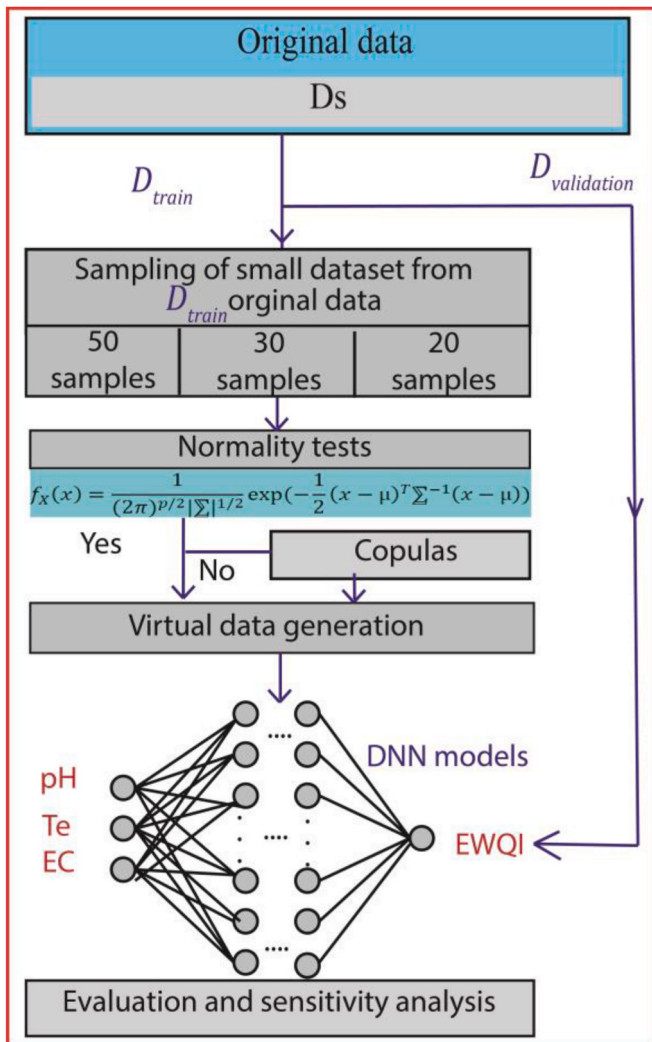


Fig. 2. Flowchart of the proposed methodology.

2.3.5. Evaluation metrics

In this study, the Coefficient of Correlation (R) (in $[-1, 1]$), Nash Sutcliffe-Efficiency (NSE) (in $]-\infty, 1]$) (Nash and Sutcliffe, 1970), Root Mean Square Error ($RMSE$) (in $[0, +\infty,]$), Mean Absolute Percentage Error ($MAPE$) (in $[0, +\infty,]$), Mean Absolute Error (MAE) (in $[0, +\infty,]$), and Percentage Bias ($PBIAS$) were considered to evaluate the developed models, as they are commonly used metric in water quality modeling (Abba et al., 2021a, 2022). Whereas, the Error Improving rate (EIR) (Chen et al., 2017) metric was used to evaluate the proposed methodology in overcoming small dataset issues. However, the computing equations are as follow:

$$NSE = 1 - \frac{\sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (P_i - \bar{O})^2} \quad (14)$$

$$R = \left(\frac{\sum_{i=1}^n (O_i - \bar{O})(P_i - \bar{P})}{[\sum_{i=1}^n (O_i - \bar{O})^2 \sum_{i=1}^n (P_i - \bar{P})^2]^{0.5}} \right) \quad (15)$$

$$RMSE = \sqrt{\frac{\sum (P_i - O_i)^2}{n}} \quad (16)$$

$$MAPE = \frac{100}{n} \sum \frac{|P_i - O_i|}{P_i} \quad (17)$$

$$MAE = \frac{1}{n} \sum |P_i - O_i| \quad (18)$$

$$EIR = \frac{MAPE_{before} - MAPE_{after}}{MAPE_{before}} 100 \quad (19)$$

O_i and P_i are observed and the model predicted values, respectively, $\bar{O}$ represents the mean of observed values, and n represents the observation instances used.

3. Results

3.1. Exploratory data analysis and feature selections

The preprocessing results of raw data showed that 720 out of 750 samples have a CBE of less than 5%. Then, exploratory data analysis was carried out to measure the MVN test, evaluate the relationship between the parameters, and the importance of features.

To evaluate whether the virtual samples can be generated according to MVN distribution, the normality measurement is required. To achieve this goal, a logarithm transformation of all variables was carried out to reduce the dimension of variables. Then, the test of Mardia (1970) shows that data are Multivariate normal with multivariate Skewness and multivariate kurtosis ($gp1 = 3.0797$, $p\text{-value} = 0.1811$) and ($gp2 = 26.5377$, $p\text{-value} = 0.2021$), respectively. Meanwhile, according to the Royston (1992) test, data are not multivariate with H-test and p-values of 25.3251 and 0.00047, respectively. However, univariate analysis is useful to further measure the normality (Kass et al., 2014). Hence, normal Q-Q plots of all variables were plotted and presented in Fig. 3. This figure shows that the univariate normality test of most variables is influenced by departures of some outliers values, which may influence Royston (1992) test results. Instead, removing 20 outlier values using the Interquartile range (IQR) showed acceptance of the MVN assumption according to the Royston test with H-test and p-values, 24.5417 and 0.004, respectively.

After data cleaning, the EWQI was recalculated then the statistical analysis was conducted. Fig. 4 illustrates the correlation matrix and the variables distribution of the cleaned dataset (700 samples). As presented in Fig. 4b, the distribution of the dataset retained was illustrated through Violin with a box plot as presented. From these figures, it was observed the variables were evenly distributed. Fig. 3a shows that the Electrical Conductivity (EC) is fairly correlated with Mg^{2+} , Ca^{2+} , Na^{2+} , SO_4^{2-} , and

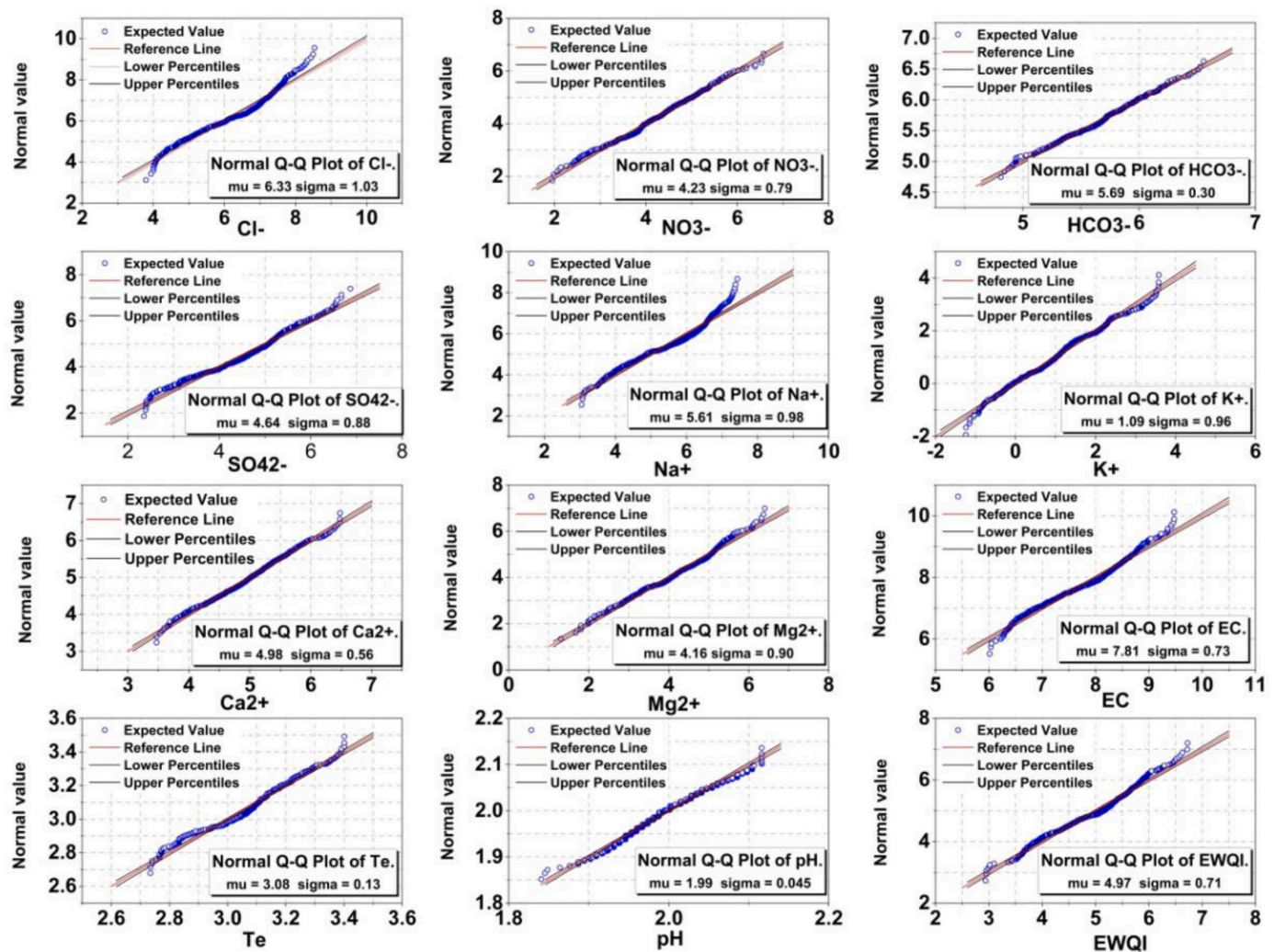


Fig. 3. Normal Q-Q plots of the logarithm transformation of datasets.

Cl^{-1} concentrations with a coefficient of correlation ranging from 0.52 to 0.96. This indicates that the last parameters are redundant variables (dependent variables) in predicting Entropy Weighted water Quality Index (EWQI). Therefore, as suggested by Kuhn and Johnson (2019) in selecting the features for training ML models, these parameters should not be selected as input variables to improve the generalization ability of the models. Meanwhile, the Te and pH are not highly correlated with the other parameters. As for the K^{+} , NO_3^{-} , and HCO_3^{-} parameters, despite being independent variables their use will reduce the practical implication of the model to be trained because their getting needs a laboratory analysis. Interestingly, using only EC, pH, and Te could be justified by the fact that they are measureable by sensor technologies in real-time. Therefore, we consider these parameters for predicting EWQI to encourage the field application of the DNN.

Multivariate normality tests of datasets 400 and 50 samples according to the Royston test were acceptable with H-test and p-values of about 21.6145, 0.005, 20.1754, and 0.006 sequentially. Hence, MVN using covariance matrixes is useful to generate virtual data. Meanwhile, Gaussian copulas with Kendall's rank correlations (Kendall, 1938) were used to generate virtual datasets.

3.2. ANN and DNN models with small data problem

In small dataset circumstances, a traditional ML model, such as a simple ANN with one hidden layer is suggested rather than DNN. In this

study, we studied three small original samples to generate virtual datasets, namely 50, 30, and 20 samples randomly sampled without replacement. ANN models were trained using the original small dataset and validated with 300 samples and compared with the DNN model developed with full original samples (400 samples).

Table 2 presents ANN model performances during the validation phase. Except for PBIAS indices, all other indices decreased in the decrease of data sizes used in the training phase. Indeed, the DNN developed (4 hidden layers with 11 neurons in each one) with 400 samples has good performances in simulating EWQI of 300 samples with $NSE = 0.88$ and $R = 0.94$. Meanwhile, according to Moriasi et al. (2007) suggestions in selecting models in water resource studies, developed ANN models (12 neurons in hidden layer) with 30 and 20 original samples are not suitable for simulation purposes, as they have $NSE < 0.50$. Such results demonstrate how much the original data sizes impact the model performances. Since data availability is a great challenge in applying DNN models, a VSG can be valuable to overcome the data shortage in predicting EWQI.

3.3. Evaluation of the proposed methodology

To evaluate the MVD-VGS method, virtual datasets with many samples ranging from 500 to 10,000 were generated from 400, 50, 30, and 20 original samples. Then 11 DNN models were trained and validated using virtual samples and 300 original samples, sequentially. A

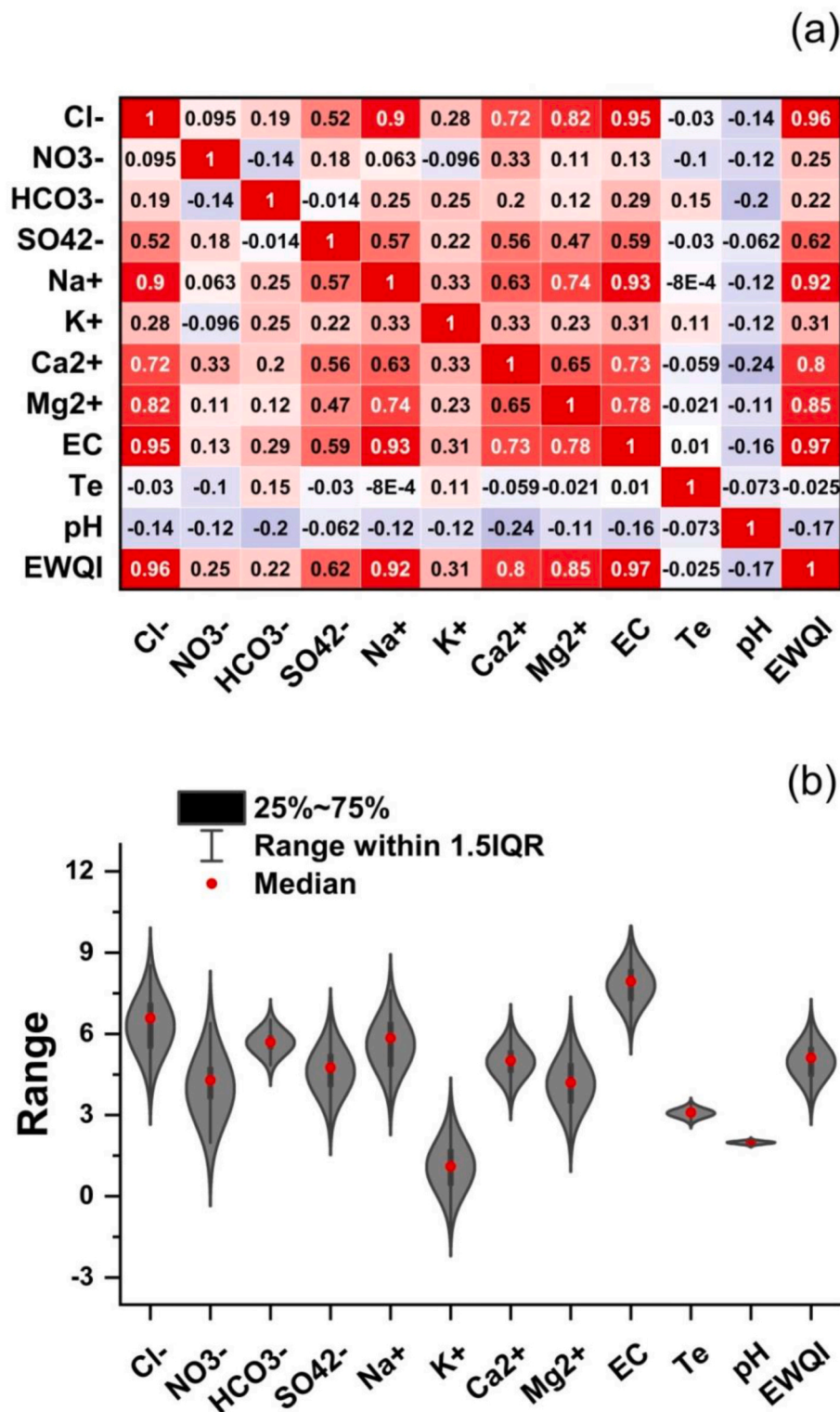


Fig. 4. Exploratory data analysis. (a): Correlation matrix between all variables, (b): Violin with box plot of the variables.

tuning process was carried out and the DNN structures with 5 hidden layer and 9 neurons were adopted using trial-error procedure method. While, the learning rates (batch size) and the epoch numbers are ranged from 0.001 to 0.01 and from 200 to 1000, respectively. Then, all models were compared with the baseline models (with only original samples).

Table 3 presents the maximum, mean, and minimum performances during the validation phase using 300 experimental data of developed DNN models with virtual datasets. It is observed a significant

improvement in the prediction accuracy of the DNN models with virtual datasets compared to the baseline model performances presented in Table 2, especially for small original datasets of 50, 30, and 20 samples. For instance, the minimum of *NSE* provided by DNN models is higher than those provided by ANN with small datasets of 50, 30, and 20 samples. As for 400 original samples, a slight improvement in DNN model performances was provided by the virtual datasets because there was a tiny discrepancy in model performances.

Table 2

Model performances with original samples.

Indice	400 original Samples	50 original Samples	30 original Samples	20 original Samples
<i>R</i>	0.94	0.78	0.78	0.72
<i>RMSE</i>	46.01	89.12	98.99	113.36
<i>NSE</i>	0.88	0.53	0.42	0.24
<i>PBIAS</i> (%)	4	4	4	4
<i>MAE</i>	29.34	63.92	59.28	90.27
<i>MAPE</i> (%)	25.75	64.36	53.69	102.39

Fig. 5 displays radar plots of various DNN model performances trained by virtual datasets, their ensemble DNN models, and those of the ANN trained with small original datasets. Generally speaking, DNN models developed by virtual datasets are more accurate than those trained by small original datasets. Moreover, it was observed that DNN model performances depend on virtual data sizes generated and their original sample size. More importantly, prediction accuracy was enhanced by the increase in the size of the original sample.

Furthermore, Error Improvement Rates of DNN models were calculated for each original small sample. Fig. 6 presents EIR distributions. From this figure, it was observed that the *EIR* highly depended on original sample sizes. Virtual datasets generated from 400 provided

lower *EIR*, as the baseline model developed by this data was more accurate than those developed by 50, 30, and 20 samples, and because the *EIR* is relative to the baseline models.

However, the best DNN models to predict the EWQI were those developed by 10,000, 4000, 6000, and 4000 virtual datasets generated from 400, 50, 30, and 20 original samples, respectively. Fig. 7 displays graphical comparisons of predicted and measured EWQI. This figure shows a good agreement between predicted and simulated values of EWQI, except for a few values. Furthermore, errors distributions showed that they were evenly distributed over zero (Fig. 7 violin plots). Thus, intervals of 25%–50% of errors are relatively narrow and most median values of the model errors are closer to zero. Therefore, it can be concluded that the errors fairly obey a normal distribution, indicating the stability of the best DNN models during the validation process.

To evaluate the ensemble DNN models (ENS-DNN), we combined the developed DNN models using the ensemble averaging method to predict the EWQI. As presented in Table 3 and Fig. 5, ENS-DNN models provided excellent performances according to model ranking suggestions of Moriasi et al. (2007). However, Fig. 8 highlights radar plots of ENS-DNN model performances. This figure confirmed that the ENS-DNN model performances depend on the size of the original samples. Despite this dependence, the comparison showed a tiny difference in performances. Therefore, the ENS-DNN is useful to improve the prediction accuracy by combining the DDN developed by virtual datasets.

Table 3

DNN model performances developed using virtual dataset generated from different original sample sizes.

Indice	400 original samples				50 Original samples				30 original samples				20 original samples			
	Max	Mean	Min	ENS-DNN	Max	Mean	Min	ENS-DNN	Max	Mean	Min	ENS-DNN	Max	Mean	Min	ENS-DNN
<i>R</i>	0.96	0.95	0.93	0.96	0.95	0.94	0.89	0.96	0.95	0.92	0.83	0.95	0.95	0.92	0.82	0.95
<i>RMSE</i>	46.89	39.70	36.69	37.79	61.88	45.84	39.73	38.63	77.42	51.23	39.97	40.88	81.71	52.50	42.92	44.00
<i>NSE</i>	0.92	0.91	0.87	0.92	0.91	0.87	0.77	0.91	0.91	0.84	0.65	0.90	0.89	0.83	0.61	0.89
<i>PBIAS</i>	3.0	2.2	0.6	0.06	3.5	2.2	0.9	0.9	5.8	1.7	0.2	1.5	10.3	6.7	4.0	6.7
<i>MAE</i>	24.77	20.93	18.83	19.30	31.38	24.26	20.48	20.74	36.66	28.06	21.55	23.31	43.41	30.09	26.93	26.37
<i>MAPE</i>	22.73	15.16	11.64	13.58	31.69	19.81	14.49	16.32	30.94	24.42	16.00	19.12	37.06	24.87	20.28	19.90

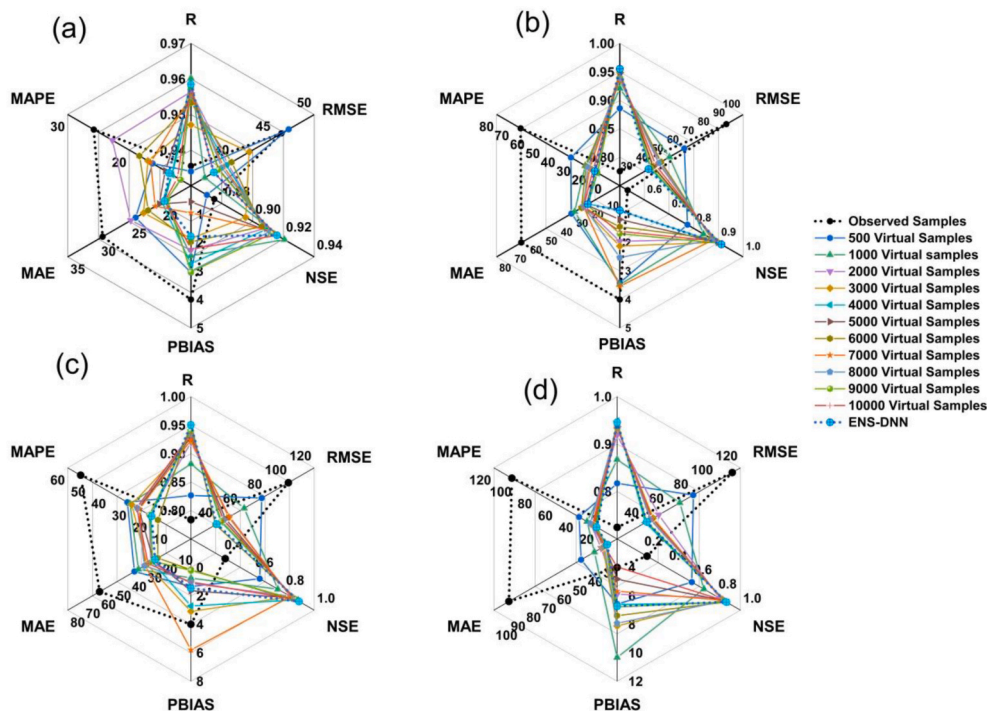


Fig. 5. Radar plots for comparing the model performances (PBIAS in absolute value). (a), (b), (c), and (d) are DNN models developed based on 400, 50, 30, and 20 observed samples in generating virtual dataset, respectively.

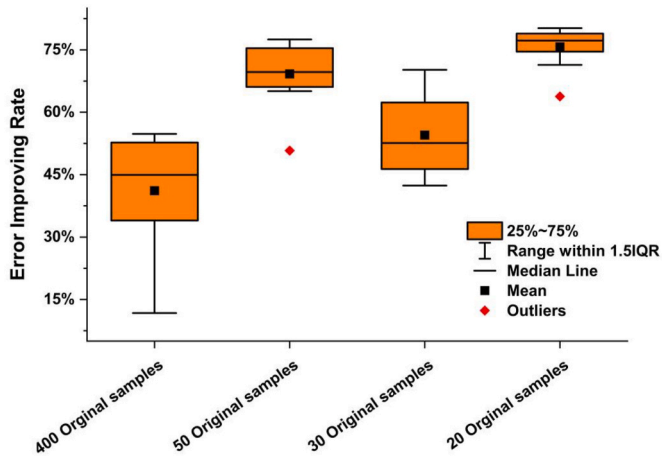


Fig. 6. Distribution of the error improving rate (EIR) for different original samples.

3.4. Sensitivity to virtual data and original sample sizes

The proposed methodology in the current study enhances the prediction accuracy of the DNN models even with small datasets. However, the appropriate amount of virtual datasets is a great challenge in developing and applying VSG methods. Indeed, complex DNN structures with big virtual data can dramatically increase the cost of calculation. The DNN structures are generally evaluated using trial-error procedures (Xu et al., 2020). Meanwhile, the sensitivity of DNN models to virtual dataset sizes should be further analyzed to evaluate the optimal virtual data sizes. In this study, we used *RMSE* and *NSE* to analyze the sensitivity of DNN models.

Accordingly, *RMSE* values decreased with the increase in virtual data

sizes (Figs. 5 and 9). However, it was observed that this decrease can be modeled by a first exponential decay (ExpDec1) according to the following equation:

$$RMSE = y_0 + A_1 \exp\left(-\frac{N_v}{t_1}\right) \quad (20)$$

A_1 , y_0 (limit), and t_1 are the ExpDec1 parameters. N_v represents the virtual data size.

Fig. 9 illustrates Scatter plots of *RMSE* against virtual data sizes, the fitted curves for each original sample, and the ExpDec1 parameters. As presented in this figure, *RMSE* values of the DNN models against N_v are fairly fitted according to the first-order exponential decay with a coefficient of determination R^2 ranging from 0.75 to 0.97. The *RMSE* limits for high N_v values are 39, 43, 46, and 46.1 for original samples of 400, 50, and 30 and samples, respectively. These limits were reached when N_v become higher than 600, 3500, and 5000, 5100 virtual samples generated, respectively, from 400, 50, 30, and 20 original samples. The virtual samples corresponding to *RMSE* limits could be considered optimal virtual data sizes because beyond these sizes DNN model performances were not significantly improved.

As for the *NSE* metric, it was observed that *NSE* increase with the increase in virtual data sizes (Figs. 5 and 10). Also, it was observed that this trend can be modeled by the Logistic function according to the following equation:

$$NSE = A_2 + \frac{A_1 - A_2}{1 + \left(\frac{N_v}{x_0}\right)^p} \quad (21)$$

A_1 , A_2 (limit), x_0 , and p represent the Logistic function parameters.

Fig. 10 displays the scatter plot of *NSE* against virtual data sizes, logistic fit curves, and logistic parameters. Similarly, the logistic function is fairly fitted the *NSE* in function to the virtual data sizes N_v with a coefficient of determination R^2 ranging from 0.78 to 0.98. Besides, the limits of *NSE* are 0.91, 0.89, 0.88, and 0.87 for original samples of 400,

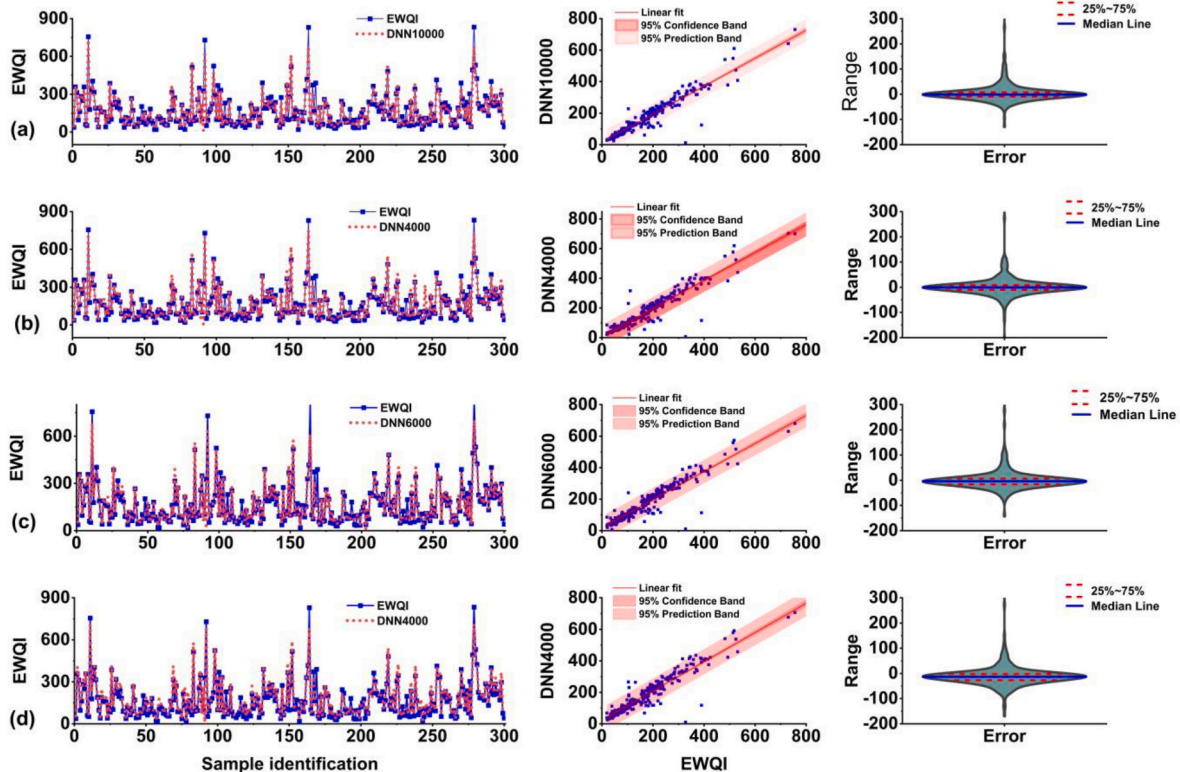


Fig. 7. Comparison of predicted and measured EWQI values using superimposed of measured EWQI and predicted values, scatter plot, and error distributions. (a), (b), (c), and, (d) are related to models developed by virtual dataset generated from 400, 50, 30, 20 original samples, respectively.

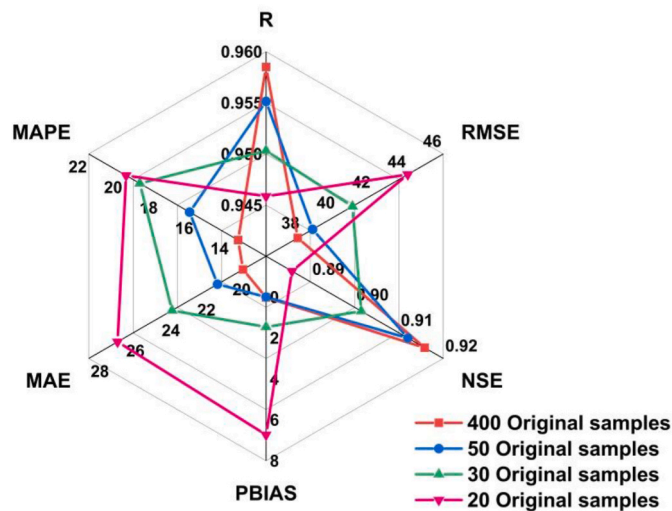


Fig. 8. Radar graphs for comparing Ensemble Deep Neural Network (ENS-DNN) models.

50, and 30 and samples, respectively. Importantly, these limits correspond to the same optimal virtual data sizes recorded in analyzing RMSE against N_v .

Determining optimal virtual data size is crucial in VSG methods as generating and using too big virtual data is not always useful. This can be justified by generating redundant variables combinations that will not improve the training process. Indeed, several authors reported that a big virtual dataset can deteriorate the prediction accuracy of the models, e.g. (De Clercq et al., 2020; El Bilali et al., 2021a; MacAllister et al., 2020; Xu et al., 2020). In this study, the optimal virtual data sizes highly depend on original sample sizes. Interestingly, Fig. 11 shows that the optimal virtual data decrease exponentially in the increase with the original sample sizes. Similarly, the limits of RMSE and NSE metric performances of the DNN models were exponentially dependent on the

original samples. This is explained by the fact that the virtual dataset generated filled up the data scarcity regions noticed in the original samples differently. Accordingly, an increase in original sample sizes decreases the data scarcity region and, therefore, it is likely to fill up appropriately using virtual data generated. However, Fig. 11 could be useful to evaluate the original sample size required and corresponding optimal virtual data sizes for given DNN mode performances to predict EWQI in small dataset circumstances.

4. Discussion

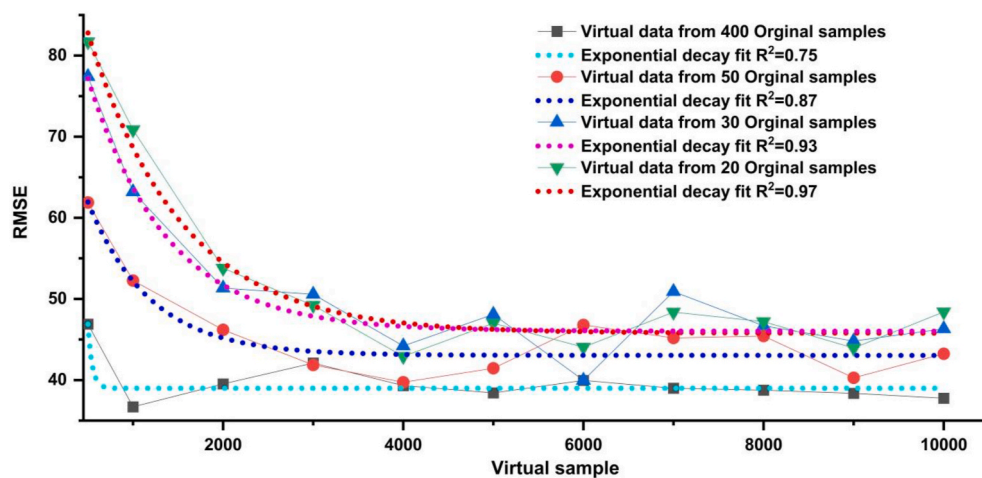
4.1. Reproducibility and practical implications

DNN is showed being an accurate and reliable approach for prediction and simulation purposes with sufficient datasets. However, data shortage either in terms of reliability or data sizes is the main impediment to applying DNN models. This constraint is generally due to funding sources to get sufficient data or insufficient monitoring data for new systems to be modeled. In this study, the proposed framework provided new insights to improve the prediction accuracy of DNN using virtual datasets generated from small datasets. The validation process with 300 experimental samples showed the robustness of the proposed methodology in predicting EWQI through DNN models developed by virtual datasets generated from small original samples.

Furthermore, the sensitivity analysis of the developed DNN models was conducted and showed that the limits of model performances and their corresponding optimal virtual data sizes were sensitive to original samples. Besides, the ENS-DNN model is suggested to improve stability and accuracy, particularly when the original sample is too small.

Nowadays, the implementation of innovative environmental technologies based on ML models requires using only real-time measurable input variables, such as the Internet of Things (IoT) (Chowdury et al., 2019; Liu et al., 2019). In this study, the EC, Temperature, and pH were used as input variables that increase the level of field application in developing IoT-based technologies for predicting and managing EWQI in real-time.

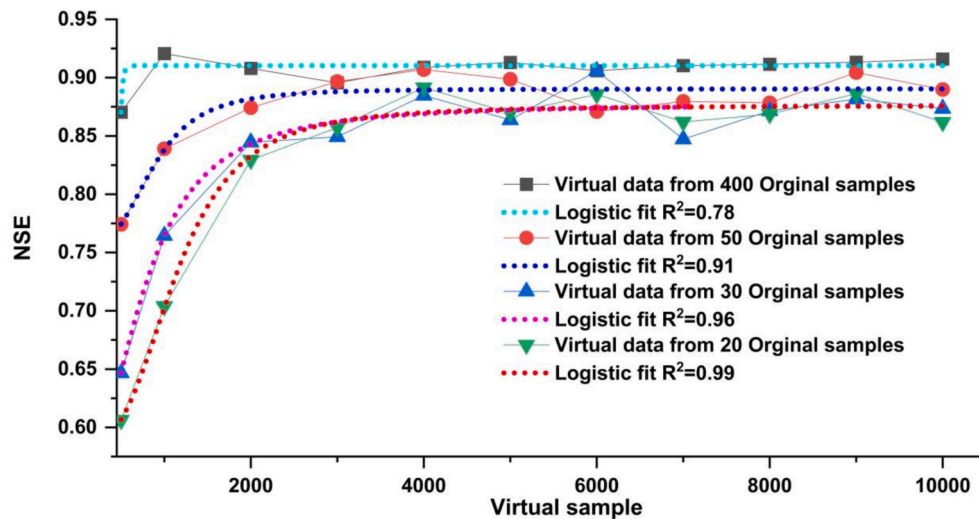
Globally, the proposed methodology is reproducible for predicting



Exponential decay function parameters

Model	ExpDec1	ExpDec1	ExpDec1	ExpDec1
Plot	Virtual data from 400 Original samples	Virtual data from 50 Original samples	Virtual data from 30 Original samples	Virtual data from 20 Original samples
y0	39	43	46	46
A1	326817	39	56	60
t1	47	689	882	1037
Reduced Chi-Sqr	2.31906	6.28657	10.00692	5.81843
R-Square (COD)	0.75408	0.87749	0.92734	0.96962
Adj. R-Square	0.6926	0.84686	0.90917	0.96202

Fig. 9. Sensitivity analysis of the DNN models to the virtual data sizes using Root Mean Square Error (RMSE).



Logistic function parameters

Model	Logistic	Logistic	Logistic	Logistic
Plot	Virtual data from 400 Original samples	Virtual data from 50 Original samples	Virtual data from 30 Original samples	Virtual data from 20 Original samples
A1	0.3587	0.7542	0.5495	0.5853
A2	0.9103	0.8901	0.8772	0.8758
x0	452	862	744	1134
p	25	3.2	3.18	3.1
Reduced Chi-Sqr	5.73604E-5	1.91941E-4	3.32558E-4	1.6328E-4
R-Square (COD)	0.7832	0.90814	0.95689	0.98603
Adj. R-Square	0.69028	0.86877	0.93842	0.98004

Fig. 10. Sensitivity analysis of the DNN models to the virtual data sizes using NSE.

EWQI either in poorly monitored zones or for new wells. Nonetheless, its application to real cases needs the determination of the minimum of the original sample to reach the required DNN performance. Hence, this study provided empirical formulas and graphs to evaluate the original samples required and to determine optimal virtual dataset sizes for reaching given performances in reproducing this methodology in other sites. Especially, the parameters of equations 20 and 21 presented in Figs. 9 and 10 are useful to cheaply assess the limits of DNN performances, such as *RMSE* and *NSE*. Also, Fig. 11 could be used to evaluate optimal virtual data sizes for each original sample.

4.2. Perspectives for developing interpretable DNN models

The DNN approach is commonly considered a black-box method because it is not explored enough in interpreting the physical processes involved in the system to be modeled. For example, in this study, although the developed DNN models are capable to predict water quality with high accuracy, they are not capable to simulate aquifer contamination processes. However, recently, some studies have been conducted and demonstrated potential capabilities of DNN and ML models in interpretation processes by incorporating physical parameters as variables (De Clercq et al., 2020; Dikshit and Pradhan, 2021; Pyo et al., 2021; Xu et al., 2022; Yokoo et al., 2022).

Interestingly, developing interpretable and accurate DNN models could be valuable to managing and planning water resources, as they will be able to simulate best practice management (BPM) actions, which are linked to physical parameters. For instance, simulating BPM actions to reduce pollution or erosion at watershed scales requires embedding physical patterns in the training of DNN models. Since it is not easy to get several instances of these parameters for a given system to be modeled, VSG methods could be useful to overcome this challenge. The proposed MVD-VSG is capable to keep the interdependences between generated variables and, therefore, the conservation of the physical

information. Consequently, it is suggested to develop interpretable DNN models exploring the MVD-VSG method to enhance the data size. Likewise, integrating process-based models as a proxy in VSG methods appeared to be suggested for developing interpretable DNN models.

5. Conclusions and future works

In this study, a new virtual sample generation method called MVD-VSG based on Multivariate Distribution is developed and successfully evaluated to enhance the DNN model performances with small datasets for predicting water quality. In MVD-VSG, the multivariate normal distribution and elliptical copulas are used to generate appropriate virtual combinations of the variables. The validation results with 300 observations showed significant enhancement of the DNN models compared to the baseline models developed only by raw original dataset. The main findings are summarized as follows:

1. Developing accurate DNN models using only 20 original samples is feasible through virtual data generated following the MVD-VSG method for predicting EWQI;
2. There are performance limits (*RMSE* and *NSE*) corresponding to optimal virtual data beyond the DNN models cannot be improved;
3. The optimal virtual data and performance limits exponentially depend on the original samples.

To sum up, the proposed methodology is powerful approach the overcome the data shortage and improve the application of DNN in various environmental fields. Moreover, its reproducibility in real engineering cases is feasible through determination of the required original samples and optimal virtual data size to reaching acceptable performances.

However, the MVD-VSG method is evaluated only for DNN models in predicting water quality. Other outstanding and accurate models are

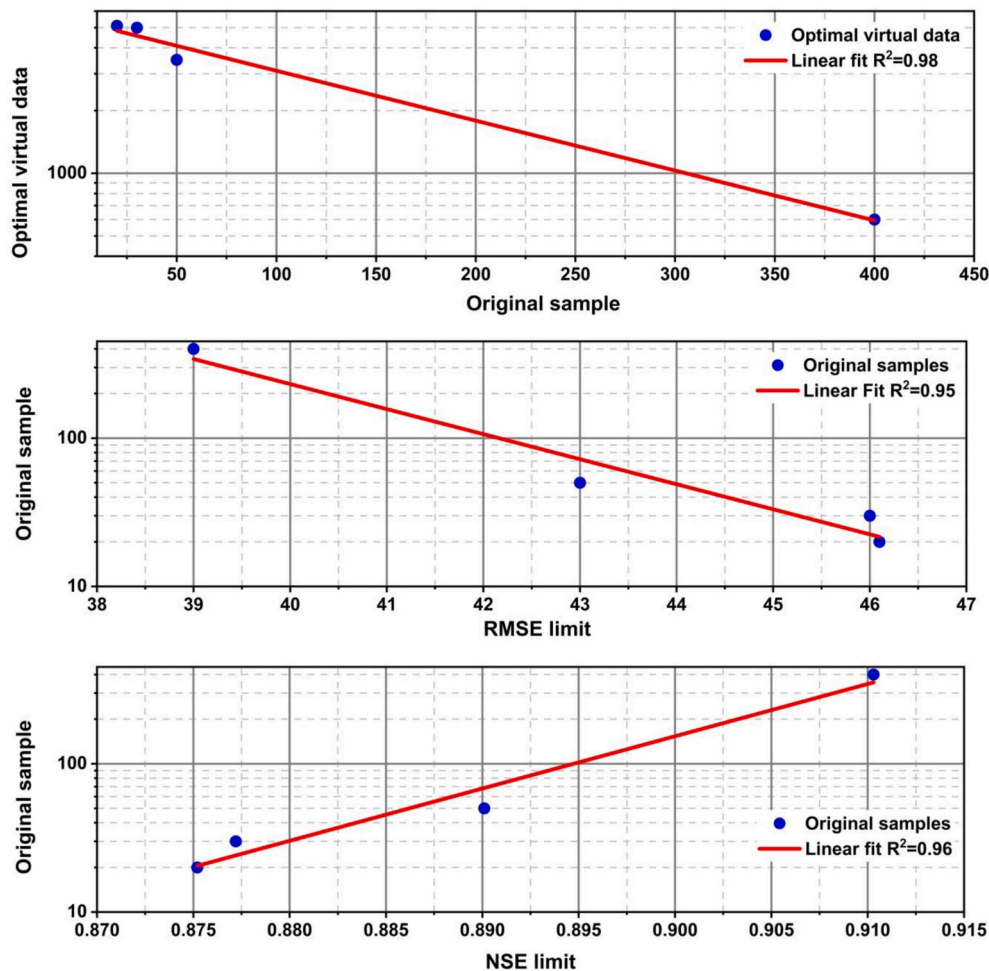


Fig. 11. Semi-Log₁₀ graph with a logarithm on y-axis of the limits of NSE and RMSE, and the optimal virtual data against original sample size.

suggested to be embedded in this method for future works, such as Conventional Neural Network, Extreme Learning Machine, and Deep forest. Similarly, other environmental fields are suggested to be explored to improve the application of data-based models even with small data. Besides, it is notable that the MVD-VSG method is evaluated for numerical input variables; its application for mixed input variables (categories + numerical input variables) is questionable. Therefore, new VSG methods should be developed and evaluated in such cases.

CRediT authorship contribution statement

Ali El Bilali: Writing – original draft, Conceptualization, Methodology, Visualization, Formal analysis, Software. **Houda Lamine:** Writing – review & editing, Formal analysis. **Abdeslam Taleb:** Supervision, Writing – review & editing, Formal analysis. **Ayoub Nafii:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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